

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Artificial Intelligence Application Development and Jira

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Class: TYIT	Batch: 1
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PRACTICAL 7

AIM :- Machine Learning

4) Implement k-fold cross-validation on a classification model.

CODE :-

```
import pandas as pd

from sklearn.model_selection import cross_val_score

from sklearn.neighbors import KNeighborsClassifier

data = pd.read_csv("iris.csv")

iris = load_iris()

X = iris.data

y = iris.target

model = KNeighborsClassifier()

scores = cross_val_score(model, X, y, cv=5)

print("Accuracy scores for each fold :")

print(scores)

print("Average Accuracy : ", round(scores.mean() * 100, 2), "%")

print("Saqlain Khan T013")
```

OUTPUT :-

```
Accuracy scores for each fold :
[0.96666667 1.          0.93333333 0.96666667 1.          ]
Average Accuracy :  97.33 %
Saqlain Khan T013
```

Different Dataset

CODE :-

```
import pandas as pd

from sklearn.model_selection import cross_val_score

from sklearn.neighbors import KNeighborsClassifier

data = pd.read_csv("housing.csv")

data = data.dropna()

data["ocean_proximity"] = data["ocean_proximity"].astype("category").cat.codes

X = data.drop("median_house_value", axis=1)

y = pd.qcut(data["median_house_value"], q=5, labels=False)

model = KNeighborsClassifier()

scores = cross_val_score(model, X, y, cv=5)

print("Accuracy scores for each fold :")

print(scores)

print("Average Accuracy :", round(scores.mean() * 100, 2), "%")

print("Saqlain Khan TO13")
```

OUTPUT :-

```
Accuracy scores for each fold :
[0.33080499 0.28284805 0.33251774 0.27655409 0.29123837]
Average Accuracy : 30.28 %
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```

5) Implement a Decision Tree classifier using the Iris or any relevant dataset.

CODE :-

```
import pandas as pd

from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.tree import DecisionTreeClassifier

data = pd.read_csv("iris.csv")
```

```

iris = load_iris()

X = iris.data

y = iris.target

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=1)

model = DecisionTreeClassifier()

model.fit(X_train, y_train)

print("Training Accuracy : ",model.score(X_train, y_train)*100)

print("Testing Accuracy : ",model.score(X_test, y_test)*100)

prediction = model.predict([X_test[0]])

print("Predicted Class : ",prediction[0])

print("Saqlain Khan TO13")

```

OUTPUT :-

```

Training Accuracy : 100.0
Testing Accuracy : 95.55555555555556
Predicted Class : 0
Saqlain Khan T013

```

Different Dataset

CODE :-

```

import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.tree import DecisionTreeRegressor

data = pd.read_csv("housing.csv")

data = pd.get_dummies(data, columns=["ocean_proximity"])

X = data.drop("median_house_value", axis=1)

y = data["median_house_value"]

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.3, random_state=1

)

```

```
model = DecisionTreeRegressor(random_state=1)

model.fit(X_train, y_train)

print("Training Score :", model.score(X_train, y_train) * 100)

print("Testing Score :", model.score(X_test, y_test) * 100)

prediction = model.predict([X_test.iloc[0]])

print("Predicted House Value :", prediction[0])

print("Saqlain Khan TO13")
```

OUTPUT :-

```
Training Score : 99.9965103862456
Testing Score : 61.17399452061585
Predicted House Value : 250000.0
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```